Particle Physics 2 Standard Model: Lecture 4 : New Revolutions in Particle Physics: Standard Model

Leonard Susskind

Transcript-derived notes curated by LazyingArt LLC

Original lecture by Leonard Susskind. Transcript-derived notes curated by LazyingArt LLC.

Chapter 1

New Revolutions in Particle Physics: The Standard Model

The lecture opens with a comic commercial break and a return to Alice and Bob, but the mathematical re-entry is an audience question: what exactly is the relation between an abstract group and the concrete matrices by which it acts on physical states? That is the real starting point of the chapter. From there the lecture proceeds in a very deliberate order: first rotations and $SU(2)$, then the parallel story for color $SU(3)$, then the way representations combine, and finally the dynamics that explains why only color-singlet states show up as free particles. These notes follow Leonard Susskind's lecture closely and were curated by LazyingArt LLC.

1.1 From Abstract Groups to Physical Representations

Susskind begins by insisting that we are *not finished* with group theory. The question from the room is exactly the right one: on the one hand a group looks like an abstract multiplication table, and on the other hand the lecture keeps talking about matrices acting on physical systems. Are these two different things, or are they really the same thing seen from two sides?

His answer is to take the cleanest possible example: rotations of space. A rotation is, first of all, an abstract operation. One can describe it without saying what it acts on. Physics then asks a second question: how does that abstract rotation act on the quantities that matter?

In classical mechanics the answer is familiar enough. A rotation acts on coordinates, or on the direction of a vector, or on the orientation of a magnet. In quantum mechanics the point becomes sharper, because the states of a system are represented by vectors in a linear vector space. The symmetry is abstract; the representation is the concrete matrix action on that vector space.

1.1.1 Question & Answer

Question. What is the difference between the abstract symmetry group and the matrices that act on states?

Answer. We should distinguish them, but we should not pull them apart. The group is the abstract symmetry itself, together with its multiplication law. A representation is a set of matrices that reproduces that same multiplication law on some chosen state space. The same rotation in space may therefore be represented by different matrices on different systems, but those matrices must still encode the same underlying group operation. In that sense the two notions are different, but in physics they are completely related.

1.2 Why Representation Dimension Depends on the System

Once the question is posed that way, the dimension of a representation stops being mysterious. It is fixed by the number of mutually orthogonal states of the system we are talking about.

If we keep only the spin of an electron, there are two such states, spin up and spin down, so rotations act by 2×2 matrices. If we had two such spins and ignored everything else, we would have four states. If we talk about a spin-one particle, there are three states, and rotations act by 3×3 matrices. If we go down to spin zero, rotations act trivially: the state is simply left alone.

In the language of the lecture,

$$\text{spin } 0: \ 1 \times 1, \qquad \text{spin } \frac{1}{2}: \ 2 \times 2, \qquad \text{spin } 1: \ 3 \times 3, \qquad \text{spin } \frac{3}{2}: \ 4 \times 4. \tag{1.1}$$

The one-by-one case is almost useless mathematically, but it is conceptually important: a spin-zero state is a trivial representation of rotations.

The point is not limited to spin. If we describe the full orbital motion of an electron, then there are infinitely many mutually orthogonal states. The amplitudes may be functions of position or momentum, and the corresponding representation of the rotation group is then infinite-dimensional. The abstract symmetry group has not changed; what has changed is the system on which it acts.

This is why the lecture spends time on what could otherwise look like old material. The same abstract symmetry operation can have many representations of different dimensions. That fact is the bridge from the familiar $SU(2)$ story of spin to the less familiar $SU(3)$ story of color.

1.3 Special Unitary Matrices and Infinitesimal Generators

We now come back, as Susskind says, to the spin of the electron and throw everything else out of our heads. For a spin-$\frac{1}{2}$ particle, rotations are represented by 2×2 unitary matrices U:

$$U^\dagger U = 1. \tag{1.2}$$

A general 2×2 complex matrix has four complex entries, hence eight real parameters. The unitary condition gives four real constraints. That gets us down to four parameters. But a rotation in ordinary space has only three independent parameters. The missing condition is

$$\det U = 1. \tag{1.3}$$

So for $SU(2)$ the counting is

$$8 - 4 - 1 = 3. \tag{1.4}$$

That is exactly the right number for spatial rotations.

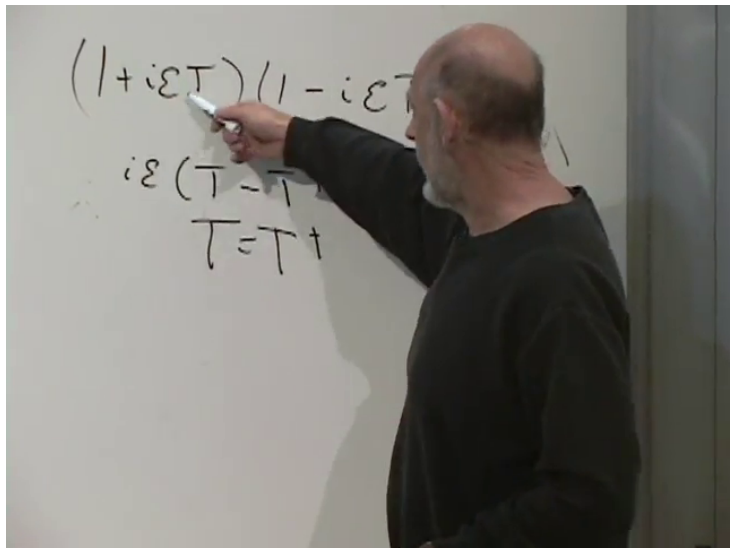

Figure 1.1: First-order unitarity implies a Hermitian generator.

The lecture pauses here to check that the determinant-one condition is consistent with multiplication. If A and B are matrices, then

$$\det(AB) = \det A \det B. \tag{1.5}$$

Also,

$$\det(U^\dagger) = (\det U)^*. \tag{1.6}$$

Taking determinants of $U^\dagger U = 1$, we find

$$(\det U)^* \det U = 1, \tag{1.7}$$

so $\det U$ must be a pure phase:

$$\det U = e^{i\theta}. \tag{1.8}$$

If we multiply U by the scalar phase $e^{-i\theta/2}$, then

$$\det\!\left(e^{-i\theta/2} U\right) = 1. \tag{1.9}$$

This isolates the special-unitary part relevant for rotations.

Here the lecture inserts an important caveat. The correspondence between spatial rotations and $SU(2)$ is not exactly one-to-one. The matrices U and $-U$ both represent the same rotation in space. So $SU(2)$ is the familiar double cover of the ordinary rotation group. The lecture does not need that fine point later, but it does not want to suppress it entirely either.

The next move is the central move of the lecture: we pass from finite group elements to infinitesimal ones. This matters because the generators are the conserved quantities. A group element very close to the identity may be written as

$$U = 1 + i\epsilon T, \tag{1.10}$$

with ϵ small. Nothing in this step is special to $SU(2)$; the same reasoning works for $SU(N)$. But here we keep the spin-$\frac{1}{2}$ picture in mind.

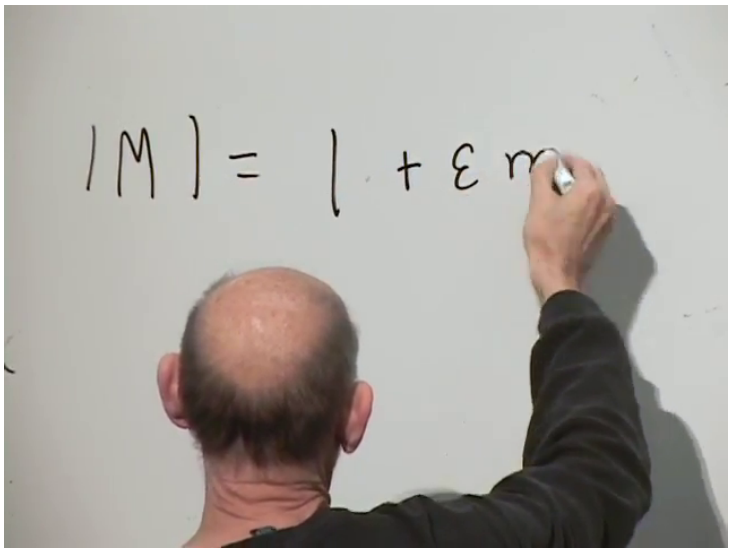

Figure 1.2: Near-identity determinant expansion on the board.

Using unitarity and keeping only terms to first order in ϵ, the board derivation is

$$(1 + i\epsilon T)(1 - i\epsilon T^\dagger) = 1, \tag{1.11}$$

$$i\epsilon\,(T - T^\dagger) = 0, \tag{1.12}$$

$$T = T^\dagger. \tag{1.13}$$

So the generator is Hermitian. That is exactly what quantum mechanics wants: Hermitian operators are observables.

At this stage the lecture asks why there are three of these generators. The reason is that the rotation group has three independent directions: infinitesimal rotations about the x-, y-, and z-axes. One can also say that the generators behave like the components of a vector. For rotations, these generators are the angular momentum operators.

Then Susskind backs up. We have used unitarity, but we have not yet used the determinant-one condition. That is where the second figure enters.

The screenshot catches the board while the determinant expansion is still being written. The fully cleaned first-order statement is

$$\det(1 + \epsilon m) = 1 + \epsilon\ \mathrm{tr}\, m + O(\epsilon^2). \tag{1.14}$$

Applying this to $U = 1 + i\epsilon T$, we get

$$\det U = 1 + i\epsilon\ \mathrm{tr}\, T + O(\epsilon^2). \tag{1.15}$$

But for $SU(2)$ we require $\det U = 1$, so the linear term must vanish:

$$\mathrm{tr}\, T = 0. \tag{1.16}$$

We have therefore arrived at the precise lecture-level characterization of the generators of $SU(2)$: they are the traceless Hermitian 2×2 matrices. There are three independent ones. In the standard

language these are the Pauli matrices, and physically they are the angular momentum operators. The lecture briefly reminds us that the whole structure of the group is encoded in the generators and, more deeply, in their commutation relations. But for present purposes the key point is simpler: the conserved quantities associated with the symmetry are the generators.

1.4 From $SU(2)$ to $SU(3)$

At this point the lecture says, half-jokingly, that we will torture Alice a little more and go on to $SU(3)$. The mathematics is intentionally repetitive: same story, bigger matrices, different physics.

A general 3×3 complex matrix has nine complex entries, hence eighteen real parameters. Unitarity gives nine real constraints, and the determinant-one condition gives one more. So

$$18 - 9 - 1 = 8. \tag{1.17}$$

Equivalently, $SU(3)$ has eight linearly independent traceless Hermitian 3×3 generators. These are the Gell-Mann matrices.

The physical interpretation now changes. A quark carries three color amplitudes, and we write its state schematically as

$$q = \begin{pmatrix} q_r \\ q_b \\ q_g \end{pmatrix}. \tag{1.18}$$

The lecture momentarily stumbles in the color names, but the stable convention is red, blue, and green. A color transformation acts by a 3×3 matrix:

$$q' = Uq. \tag{1.19}$$

Thus color rotations mix red, blue, and green exactly as spin rotations mix spin-up and spin-down amplitudes.

A pure red quark is, in that sense, the color analogue of a pure spin-up state. After an $SU(3)$ transformation it becomes a quantum superposition of red, blue, and green.

Now we ask the same question for anti-quarks. Their field operators are the complex conjugates of the quark operators, so they transform in the complex-conjugate representation:

$$\bar{q}' = U^* \bar{q}. \tag{1.20}$$

This is the $\bar{3}$, the anti-triplet. The lecture uses the $U(1)$ example to make the point intuitive: if an electron wavefunction transforms by $e^{i\theta}$, then the positron transforms by $e^{-i\theta}$. They are complex-conjugate representations. The same idea is now being used for color.

For $SU(2)$ the complex-conjugate representation is equivalent to the original one; that is a special feature of $SU(2)$. For $SU(3)$, however, the 3 and the $\bar{3}$ are distinct. The lecture states, at its own level of precision, that the generators in the $\bar{3}$ representation are the negatives of those in the 3 representation. That is the sense in which an anti-quark carries the opposite color.

1.5 How Representations Combine

Before doing color products, Susskind steps back again and asks how representations combine at all. This is not a decorative review. Without it, formulas like $3 \otimes 3 = \bar{3} \oplus 6$ would just look like mysterious multiplication tables.

Take two spin-$\frac{1}{2}$ systems and ignore all orbital motion. If we put the systems together, we get four basis states:

$$\uparrow\uparrow, \qquad \uparrow\downarrow, \qquad \downarrow\uparrow, \qquad \downarrow\downarrow\,.$$

In operator language we can think of them as built from an object with two indices, $\psi_i \phi_j$, where i and j each take the values $\uparrow$ or $\downarrow$. The $SU(2)$ action on ψ and on ϕ induces an action on this four-dimensional composite space.

The key discovery is that one particular linear combination does not mix with the others:

$$\epsilon_{ij}\psi_i\phi_j = \psi_\uparrow\phi_\downarrow - \psi_\downarrow\phi_\uparrow. \tag{1.21}$$

This is the singlet. The remaining three states mix among themselves:

$$\psi_\uparrow\phi_\uparrow, \qquad \psi_\downarrow\phi_\downarrow, \qquad \psi_\uparrow\phi_\downarrow + \psi_\downarrow\phi_\uparrow. \tag{1.22}$$

They form the triplet. Thus

$$2 \otimes 2 = 1 \oplus 3. \tag{1.23}$$

The lecture does not insist on the $1/\sqrt{2}$ normalizations, so we follow that convention here.

This is just the familiar statement that two spin-$\frac{1}{2}$ objects may combine into spin zero or spin one. The lecture even reminds us of the old atomic-language example: ortho- and para-hydrogen.

1.5.1 Question & Answer

Question. What do equations like $3 \otimes 3 = \bar{3} \oplus 6$ actually mean?

Answer. They mean that the composite state space is reducible. The left-hand side is the full tensor-product space, with all the states obtained by combining the two systems. The right-hand side says that in a suitable basis the group action splits into invariant blocks. In the $2 \otimes 2$ example, one can choose the basis consisting of the singlet and the three triplet states; then the 4×4 matrices representing the group become block diagonal, with a 1×1 block and a 3×3 block. Exactly the same logic lies behind the color formulas. The plus sign on the board is not arithmetic addition; in clean notes we write it as $\oplus$, decomposition into irreducible pieces.

With that warm-up in place, the color products begin to make sense.

For two quarks, the lecture states

$$3 \otimes 3 = \bar{3} \oplus 6. \tag{1.24}$$

The interpretation is simple and important. Two quarks give $3 \times 3 = 9$ color states. If we organize those states by symmetry under interchange of the two color indices, then the antisymmetric part has dimension 3 and furnishes the $\bar{3}$, while the symmetric part has dimension 6 and furnishes the 6.

For a quark and an anti-quark we get

$$3 \otimes \bar{3} = 1 \oplus 8. \tag{1.25}$$

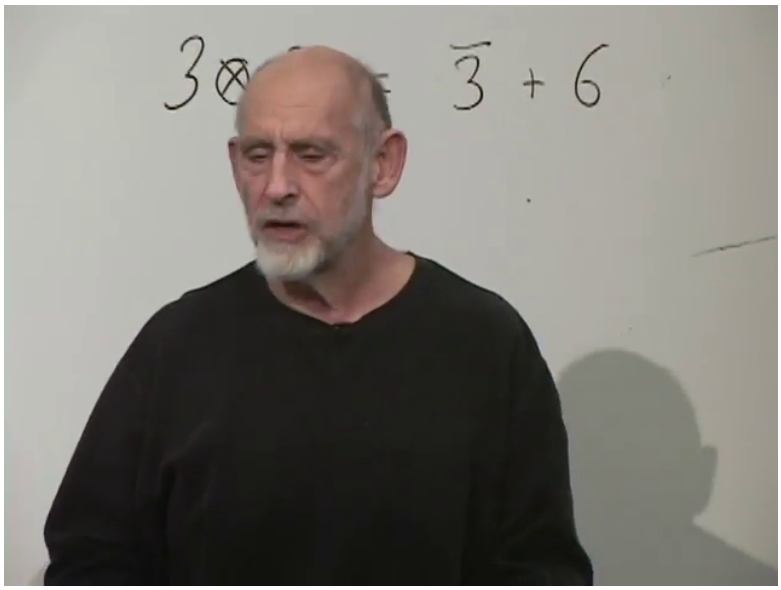

Figure 1.3: Two-quark triplet product decomposed into $\bar{3}$ and 6.

One linear combination is invariant under color:

$$q_r\bar{q}_r + q_b\bar{q}_b + q_g\bar{q}_g, \tag{1.26}$$

up to normalization. That is the singlet. The remaining eight states mix into one another and form the octet, the representation that transforms the same way as the eight generators. The lecture mentions the name *adjoint representation*, but the content is more important than the label: it is the representation carried by the gluons.

For three quarks the lecture proceeds one step at a time. First,

$$3 \otimes 3 = \bar{3} \oplus 6. \tag{1.27}$$

Then combine with the third quark. The $\bar{3} \otimes 3$ branch gives

$$\bar{3} \otimes 3 = 1 \oplus 8, \tag{1.28}$$

so

$$3 \otimes 3 \otimes 3 \supset 1. \tag{1.29}$$

The singlet is written with the antisymmetric epsilon tensor:

$$\epsilon_{ijk} q_i q_j q_k. \tag{1.30}$$

Because ϵ_{ijk} vanishes unless i, j, and k are all different, this is precisely the antisymmetrized red–green–blue combination.

The lecture briefly remarks, cautiously, that the other branch produces higher irreducible representations, for example the $6 \otimes 3$ branch leading to higher multiplets. But the singlet is the branch that matters most physically, and it is the one to which the lecture immediately returns.

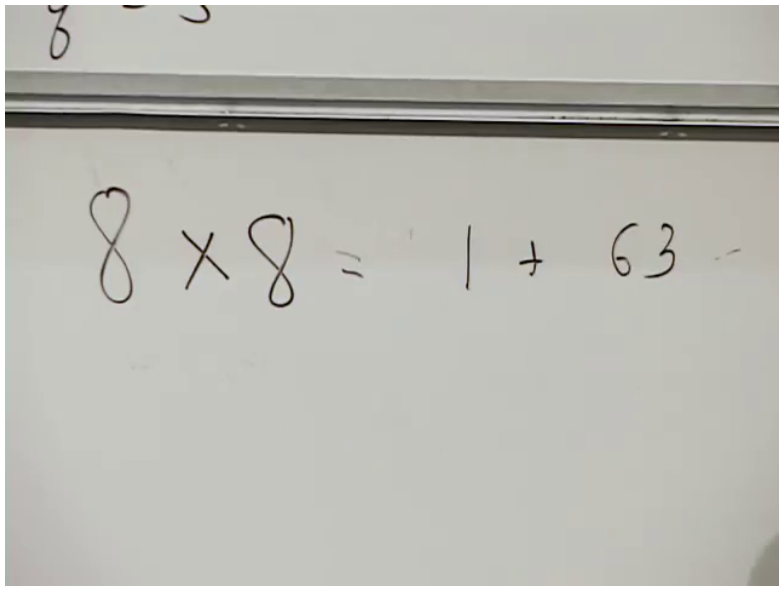

Figure 1.4: Two-gluon state count written as $8 \times 8 = 1 + 63$.

1.6 Singlets, Hadrons, and the Octet of Gluons

At this point representation theory is translated into particle language. That translation is not an afterthought; it is the whole point of the exercise.

A quark field transforms as a 3, an anti-quark field as a $\bar{3}$, and a gluon field as an 8. Of course, these symbols do not mean that particles are literally equal to numbers. They mean only the representation under which the corresponding fields transform.

The gluon deserves special emphasis. With respect to color, it behaves like a quark–anti-quark combination with the singlet removed. That is why the gluon sits in the octet rather than the singlet:

$$3 \otimes \bar{3} = 1 \oplus 8. \tag{1.31}$$

The omitted singlet does not mix with the octet under $SU(3)$, so throwing it away is mathematically consistent.

Now the lecture asks what singlets can be built from these ingredients. The basic answers are:

- a color-singlet qqq state, called a baryon,
- a color-singlet $q\bar{q}$ state, called a meson,
- a color-singlet state built from gluons, called a glueball.

The watchword is color singlets.

Two gluons provide the simplest gluonic example.

The board first records the counting checkpoint

$$8 \times 8 = 1 + 63. \tag{1.32}$$

The meaning is straightforward. Two gluons give 64 states in all; one combination is a singlet, and the remaining 63 states belong to non-singlet sectors.

Later in the discussion the lecture fills this in more explicitly. In cleaned notation one writes

$$8 \otimes 8 = 1 \oplus 8 \oplus 8 \oplus 10 \oplus \overline{10} \oplus 27. \tag{1.33}$$

This should be read cautiously: the screenshot itself shows only the state count, not the full irreducible decomposition. But the later discussion does make clear what the formula means. In an appropriate basis, the 64×64 matrices acting on the two-gluon system become block diagonal, with separate blocks for the singlet, the two octets, the 10, the $\overline{10}$, and the 27. The one-dimensional block is the glueball sector.

Susskind then sharpens the physical point. The observable hadrons are singlets. Baryons are the singlet states in $3 \otimes 3 \otimes 3$. Mesons are the singlet states in $3 \otimes \bar{3}$. Glueballs are the singlet states in products of gluon representations. All of them are colorless.

This is also why the free hadrons carry only integer electric charge. The allowed singlet combinations are built from quark–anti-quark pairs, three-quark combinations, or colorless gluonic composites. Those combinations never leave us with isolated fractional quark charge. In the lecture this is not yet the full dynamical explanation, but it is the first structural payoff of color singlets.

There is also a useful analogy here. A singlet under $SU(3)$ is the color analogue of a neutral object under $U(1)$. In electrodynamics an electron transforms by a phase $e^{i\theta}$, a positron by $e^{-i\theta}$, and the phases cancel in the neutral pair. Likewise, a color singlet is the combination on which the group action cancels out completely.

1.7 Why Only Singlets Survive

Now the lecture turns the screw. Up to this point we have classified states. But classification alone does not tell us why only singlets show up in nature as free particles. Are the other representations merely unlikely? The lecture's answer is more severe: the issue is not probability but energy.

Susskind first phrases this provocatively by saying that the energy of the non-singlet states is effectively infinite. He then explains what that means.

The story begins with Nambu. Well before quantum chromodynamics was fully established, Nambu realized that if physical states were forced to be color singlets, then fractionally charged free particles would never appear. That was the original motivation. The next question was what kind of dynamics could actually enforce such a rule.

The lecture answers by analogy with electrodynamics. A plus charge and a minus charge attract each other, and a neutral system has less field energy than a system with net charge. One way to see this is through field lines: a net charge leaves field stretching out to infinity, and that field carries energy. A neutral pair can cancel its field outside the system and therefore has lower energy. If the electromagnetic coupling were enormously larger than it actually is, isolated electric charges might themselves be effectively confined.

Nambu's idea was to transfer this logic to color. The color generators are the sources of the gluon field, just as electric charge is the source of the electromagnetic field. At the lecture's level of description, a gauge theory is precisely a theory in which conserved quantities source Maxwell-like, photon-like fields. In QCD the relevant conserved quantities are the eight color generators, and the corresponding gauge fields are the eight gluon fields.

But here the analogy with electrodynamics breaks in a crucial and fruitful way. Photons are neutral

and therefore do not source one another. Electrodynamics is linear in that sense. Gluons, by contrast, transform in the octet. They carry color. So the gluon field interacts with itself.

That is the key dynamical fact of the lecture. Because the gluon field is itself colored, its lines of flux do not spread independently the way electric field lines do in a simple dipole pattern. Instead, they attract one another and bundle into a flux tube. The field inside the tube is approximately uniform, which means that the energy per unit length is approximately uniform as well. Hence the energy of a separated quark–anti-quark pair grows roughly linearly with the separation:

$$V(r) \propto r. \tag{1.34}$$

The lecture does not derive a string tension or give a detailed potential. It gives the physical picture, and the picture is enough: color is confining because the flux does not fan out; it gets squeezed into a tube.

This immediately explains why isolated color is not seen. If we try to pull a quark away from an anti-quark inside a meson, the flux tube stretches and the energy rises. Either the quark loses its kinetic energy and is pulled back, or the tube breaks by pair creation:

$$q\bar{q} \longrightarrow (q\bar{q}) + (q\bar{q}). \tag{1.35}$$

In either case we do not get a free quark. What we get is one meson, or two mesons, or a whole cascade of hadrons. When the process repeats after a hard collision, the lecture says, a whole spray of hadrons flies out. That is a jet.

The same physical reasoning leaves room for baryons. Three quarks can combine into a singlet, and the flux lines can come together in threes. The lecture treats that as the dynamical counterpart of the earlier group-theoretic fact that $3 \otimes 3 \otimes 3$ contains a singlet.

At this point Susskind also sharpens the meaning of *gauge theory*. A gauge theory, at lecture level, is one in which conserved quantities are coupled to photon-like fields. The connection to symmetry is immediate: conserved quantities arise from symmetries. He then adds one more physical observation. If flux lines can end only on charges, then the sources are conserved. Otherwise one could destroy a charge and simultaneously alter the field arbitrarily far away, violating the basic causal structure of the theory. So conservation of charge and the behavior of flux lines are not two separate facts; they are two descriptions of the same underlying gauge structure.

This completes the QCD part of the Standard Model as it is developed in the lecture. We now have the eight generators of $SU(3)$, together with the familiar $U(1)$ case of electrodynamics, giving nine conserved quantities and nine gauge fields: the eight gluons and the photon. The remaining ingredient is the extra $SU(2)$ of the Standard Model, together with the Higgs sector, which the lecture postpones to the next meeting. In the lecture's own bookkeeping, we now have most of the working parts of

$$SU(3) \times SU(2) \times U(1), \tag{1.36}$$

but not yet the full package.

1.8 Summary

The lecture starts by reopening group theory, not by leaving it behind. Its first job is to separate the abstract symmetry group from the matrix representation without breaking the relation between

them. From there it reviews the dependence of representation dimension on the system, re-derives the $SU(2)$ conditions of unitarity and determinant one, and then shows that the infinitesimal generators are traceless Hermitian matrices, physically the angular momentum operators.

Only after that spine is secure does the lecture repeat the same pattern for $SU(3)$. Quarks live in the 3, anti-quarks in the $\bar{3}$, and gluons in the 8. The tensor-product decompositions $3 \otimes 3$, $3 \otimes \bar{3}$, $3 \otimes 3 \otimes 3$, and $8 \otimes 8$ then explain how mesons, baryons, and glueballs arise as color singlets. Finally the lecture answers the physical question that the algebra alone cannot answer: why only singlets? Because the gluon field carries color, it self-interacts, squeezes flux into tubes, and makes isolated color energetically inaccessible. That is the group theory of QCD carried over into dynamics, and it is the point from which the Standard Model can now be finished.